

ATTRIBUTION-AWARE BAYESIAN MEMORY GOVERNANCE

A Step-by-Step Explanation and Six-Month Research Guide

Detailed interpretation of the implementation scaffold v0.1 with industrial anomaly-detection examples

Item	Description
Source basis	Minimum Implementation Scaffold for Attribution-Aware Bayesian Memory Governance, Version 0.1, 30 Aug 2026.
Purpose	Explain the scaffold logically, define every major mathematical variable, and show how the pieces connect during online industrial inspection.
Architecture policy	No architectural replacement. This document explains the existing scaffold and adds only clearly marked explanatory expansions.
Primary dataset	Real-IAD, using category-level cross-fitting and a single-view protocol for the first decisive experiments.
Research horizon	Six months / approximately 24 weeks, using the scaffold's staged gate order.

How to read this guide

Read Part I first for the big idea, then Part II for the formulas, and Part III for the actual research schedule. The same illustrative example - a product with a suspicious surface/geometry pattern - is reused so that the mathematics stays connected to a real inspection process.

Part I - The Research Idea in Plain Language

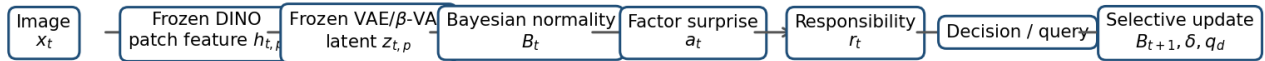
1. What changed from the first paper?

The first paper kept the visual backbone frozen and allowed only external deployment state - memory, correction, calibration, and query allocation - to change. The new study preserves that principle. The new question is more specific: when a human gives richer feedback than a binary normal/defect label, where should that information be written inside memory?

One-sentence idea

Instead of updating all of memory when a human corrects the detector, identify which latent factor groups carried the surprising evidence and update those groups more strongly.

2. The whole system before mathematics



Only queried feedback may update external state. DINO and the factorizer stay frozen during the target stream.

The representation is frozen during the target stream. Online learning happens in Bayesian normal-state statistics, source-to-factor memory, and the bounded correction state. The query controller is added only after the representation and memory mechanisms are shown to work.

3. A concrete inspection example to keep in mind

Illustrative scenario: a camera inspects a metal product. A small region looks unusual. The appearance might correspond to a harmless normal variation, a scratch-like surface anomaly, a local deformation, or another defect manifestation. The factor groups do not have to be semantic. We only ask whether recurring defect manifestations produce repeatable responsibility patterns over a small subset of latent groups.

1. DINO extracts a patch descriptor from the suspicious region.
2. A frozen VAE or beta-VAE maps that descriptor into 32 latent coordinates grouped into 8 groups.
3. The Bayesian normal model asks how surprising each factor group is relative to normal support evidence.
4. The responsibility vector assigns more credit to factor groups that are unusually surprising.
5. If a human says the item is actually normal, the most responsible factors receive most of the normal-memory update.
6. If the item is defective and a defect-source label is available, the system learns which responsibility pattern usually accompanies that human-named manifestation.
7. Only after these mechanisms are validated does the system test whether attribution uncertainty improves query allocation.

4. The five claims (C1-C5)

Claim	Question	What would support it?
C1 Factor addressability	Does beta-regularised latent organisation	Better source/aspect separability and/or

	give repeatable factor addresses?	spatial attribution than beta=1, without losing useful anomaly ranking.
C2 Factor-addressed memory	Can feedback update the responsible factors without disturbing unrelated factors?	Lower collateral posterior drift under identical queried samples.
C3 Source-enriched feedback	Does (normal/defect + defect source) teach more than a binary label alone?	Better future attribution or operating performance than binary-only and shuffled-source controls.
C4 Bayesian governance	Does posterior uncertainty help in sparse/noisy conditions?	Better calibration/stability than a matched deterministic Gaussian model.
C5 Information-aware querying	Can attribution information improve supervision efficiency?	Better performance-query frontier or fewer queries to reach a target operating point.

5. Experimental philosophy: why the stages are deliberately separated

- Fixed-query replay is used for C1-C4 so every method sees exactly the same queried samples. This isolates the mechanism being tested.
- End-to-end querying is used only for C5, after the memory mechanism is established.
- Target-category labels may not be used to train the representation or choose hyperparameters before target evaluation.
- If a simpler claim fails, later complexity should not be added to hide the failure.

Part II - Dataset, Representation, and Mathematics

6. Dataset protocol and what the simulated human reveals

Real-IAD is the principal dataset because it keeps continuity with the first paper, provides long per-category streams, pixel-level anomaly masks, and specific defect-type labels. The pixel mask is kept offline for evaluation and is not given to the online learner in the principal experiment.

Equation (1). Binary human feedback

$$h_t^{\text{bin}} = y_t$$

Symbol	Meaning
h_t^{bin}	The information returned by the simulated human under the binary condition.
y_t	Ground-truth label of item t: normal or defective.

What the equation means

A queried human returns only the operating label. The algorithm learns whether the current item is normal or defective, but receives no defect-source/aspect information.

Industrial example (illustrative)

An inspector sees a suspicious item and simply answers "defective". The system knows the decision was wrong or right, but not which visual aspect the inspector is referring to.

Equation (2). Source-enriched human feedback

$$h_t^{\text{src}} = (y_t, d_t)$$

Symbol	Meaning
h_t^{src}	Human feedback packet in the source-enriched condition.
y_t	Normal/defect label.
d_t	Recorded defect type/source-aspect label, available when $y_t=1$.

What the equation means

The human gives the same binary decision plus a compact description of the defect manifestation. The proposal deliberately treats d_t as a visual source/aspect label, not as a manufacturing root cause.

Industrial example (illustrative)

The inspector says "defective - this is a surface scratch-type manifestation" rather than only "defective". The algorithm can associate that label with a recurring latent responsibility pattern.

7. Category-level cross-fitting: preventing target leakage

8. Split the 30 Real-IAD categories into five folds of six categories.
9. Use 24 categories as source categories and six as held-out target categories.
10. Train the factorising module and estimate Bayesian prior hyperparameters only from source-category normal data.
11. Freeze the representation and prior hyperparameters.
12. Run the four-shot online FSAD protocol on the six target categories.
13. Rotate folds so every category is evaluated exactly once as a target category.

Why this matters

If target labels are used to shape the beta-VAE or the prior before the online stream, the method would no longer be testing sparse deployment-time adaptation fairly.

8. Frozen DINO patch features and factorisation

Let $h_{\{t,p\}}$ be the frozen patch descriptor for image t and patch location p . DINO remains unchanged. The factorising module operates on those descriptors, not on raw pixels.

Equation (3). Feature-space VAE / beta-VAE mapping

$$h_{t,p} \longrightarrow z_{t,p} \longrightarrow \hat{h}_{t,p}$$

Symbol	Meaning
$h_{\{t,p\}}$	Frozen DINO patch descriptor for image t, patch p.
$q_{\phi}(z h)$	VAE encoder distribution with parameters ϕ .
$z_{\{t,p\}}$	Latent code for the patch.
$p_{\psi}(h z)$	Decoder distribution with parameters ψ .
$\hat{h}_{\{t,p\}}$	Reconstructed DINO feature, not reconstructed image pixels.

What the equation means

The VAE learns a compact organisation of the already useful DINO feature space. In the proposed default, z has 32 dimensions split into 8 groups of four dimensions.

Industrial example (illustrative)

A small scratch-like patch and a normal textured patch are first represented by DINO. The VAE compresses those high-dimensional patch features into grouped latent coordinates. The groups do not need names such as "scratch" or "geometry".

Equation (4). beta-VAE objective in feature space

$$\mathcal{L}_{\beta} = \mathbb{E}_{q_{\phi}(z|h)} \left[\|h - \hat{h}\|_2^2 \right] + \beta D_{\text{KL}}(q_{\phi}(z | h) \| \mathcal{N}(0, I))$$

Symbol	Meaning
$\mathcal{L}_{\beta}$	Training loss for the factorising module.
$\ h - \hat{h}\ _2^2$	Reconstruction error in DINO feature space.
β	Strength of KL regularisation. $\beta=1$ gives the ordinary VAE control; $\beta>1$ adds stronger factorisation pressure.
D_{KL}	Kullback-Leibler divergence.
$\mathcal{N}(0, I)$	Standard normal latent prior.

What the equation means

The first term asks the latent code to retain enough information to reconstruct the DINO descriptor. The second term pressures the latent distribution toward a simpler, more factorised organisation. The study does not assume that these factors correspond to semantic or physical causes.

Industrial example (illustrative)

If beta is too small, the latent code may be an arbitrary nonlinear compression. If beta is too large, useful anomaly information may be lost. This is why R1 (beta=1) and R_beta (beta>1) must be compared directly.

Implementation note

Controls: R0 = original frozen representation, R1 = beta=1 VAE, R_beta = same VAE architecture with beta>1. The scaffold suggests beta=4 as the pre-registered default and {2,4,8} only for source-side sensitivity.

9. Bayesian normality: what is uncertain?

The Bayesian model is not placed over DINO weights or VAE weights. It models uncertainty in the normal latent distribution for each coordinate. Because each target category begins with only four normal support images, the scaffold uses a diagonal Normal-Inverse-Gamma (NIG) model.

Equation (5). Latent coordinate likelihood

$$z^{(g,j)} \mid \mu_{g,j}, \sigma_{g,j}^2 \sim \mathcal{N}(\mu_{g,j}, \sigma_{g,j}^2)$$

Symbol	Meaning
$z^{(g,j)}$	Coordinate j inside factor group g.
$\mu_{\{g,j\}}$	Unknown normal mean of that coordinate.
$\sigma_{\{g,j\}}^2$	Unknown normal variance of that coordinate.
$N(\mu, \sigma^2)$	Gaussian observation model.

What the equation means

Normal latent coordinates are assumed Gaussian conditional on their unknown mean and variance. This is intentionally simple and independently factorised across coordinates.

Industrial example (illustrative)

If factor group 3 contains four coordinates, each coordinate gets its own uncertain normal mean and variance.

Equation (6). Prior over the normal mean

$$\mu_{g,j} \mid \sigma_{g,j}^2 \sim \mathcal{N}\left(\mu_{0,g,j}, \frac{\sigma_{g,j}^2}{\kappa_{0,g,j}}\right)$$

Symbol	Meaning
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mu_0	Source-estimated prior mean.
kappa_0	Prior strength / effective precision for the mean.
sigma^2/kappa_0	Conditional uncertainty of the mean given variance.

What the equation means

The source categories provide prior knowledge about where a normal latent coordinate is likely to sit. kappa_0 says how strongly that prior should be trusted.

Equation (7). Prior over the normal variance

$$\sigma_{g,j}^2 \sim \text{InvGamma}(\alpha_{0,g,j}, \beta_{0,g,j})$$

Symbol	Meaning
alpha_0, beta_0	Source-estimated shape/scale hyperparameters of the inverse-gamma prior.
InvGamma	Conjugate prior for an unknown Gaussian variance.

What the equation means

This prior represents uncertainty about how much normal variation is expected in that coordinate. Four target supports then update the source prior into a category-specific posterior.

Explanatory expansion - One concrete fractional NIG update

$$\kappa^+ = \kappa + w, \quad \mu^+ = \frac{\kappa\mu + wz}{\kappa + w}, \quad \alpha^+ = \alpha + \frac{w}{2}, \quad \beta^+ = \beta + \frac{\kappa w(z - \mu)^2}{2(\kappa + w)}$$

Why this is included

The scaffold says the NIG sufficient statistics are updated fractionally using the factor weight alpha_{g,t}, but it does not print the conjugate update equations. The formula above is one standard implementation for a single observation z with fractional weight w. Use it as an implementation explanation, not as an additional theoretical claim.

Explanatory expansion - Posterior predictive Student-t

$$z_{\text{new}} \mid D_t \sim t_\nu(\mu_t, s_t), \quad \nu = 2\alpha_t, \quad s_t^2 = \frac{\beta_t(\kappa_t + 1)}{\alpha_t \kappa_t}$$

Why this is included

The scaffold states that the NIG posterior predictive is Student-t. This expansion shows why sparse-data uncertainty is preserved: small alpha/kappa values produce heavier-tailed predictive uncertainty instead of pretending that four supports determine the normal distribution exactly.

10. From normality to anomaly evidence

Equation (8). Patch-level factor surprise

$$a_{g,t,p} = - \sum_{j \in g} \log p\left(z_{t,p}^{(g,j)} \mid B_t\right)$$

Symbol	Meaning
$a_{\{g,t,p\}}$	Anomaly surprise of factor group g at image t , patch p .
$j \in g$	The coordinates belonging to group g .
$p(z \mid B_t)$	Posterior-predictive probability of the latent coordinate under current normal belief B_t .
$-\log$ probability	Large value means the observed latent coordinate is unlikely under normality.

What the equation means

The detector does not ask whether a factor has a semantic name. It asks whether the current latent coordinates are surprising under the learned normal distribution.

Industrial example (illustrative)

If a patch on a metal rim produces an unusual pattern mainly in group 3, group 3 receives high negative-log predictive probability.

Equation (9). Patch-to-image aggregation within each factor

$$a_{g,t} = \text{TopMean}_{p,k} a_{g,t,p}$$

Symbol	Meaning
$\text{TopMean}_{\{p,k\}}$	Average of the k largest patch surprises for that factor.
$a_{\{g,t\}}$	Image-level evidence for factor g .

What the equation means

Industrial defects are often local. Taking the top tail preserves local defect sensitivity but is less brittle than

using one single maximum patch.

Equation (10). Factor evidence vector

$$\mathbf{a}_t = (a_{1,t}, \dots, a_{G,t})$$

Symbol	Meaning
$\mathbf{a}_t$	Vector containing the image-level surprise of all G factor groups.
G	Number of factor groups; default G=8.

What the equation means

This vector is the common object used both for anomaly scoring and attribution.

Equation (11). Image anomaly score

$$s_t = \text{TopMean}_{g,k_g} a_{g,t}$$

Symbol	Meaning
s_t	Single image-level anomaly score.
$\text{TopMean}_{\{g,k_g\}}$	Aggregate the most anomalous factor groups.

What the equation means

The initial method keeps one image-level score and one operating threshold. Factor-specific thresholds are deliberately deferred.

11. Machine responsibility: where did the anomaly evidence come from?

Equation (12). Responsibility from standardised factor surprise

$$r_{g,t} = \frac{\exp(\lambda \tilde{a}_{g,t})}{\sum_{h=1}^G \exp(\lambda \tilde{a}_{h,t})}, \quad \mathbf{r}_t = (r_{1,t}, \dots, r_{G,t})$$

Symbol	Meaning
$r_{\{g,t\}}$	Responsibility assigned to factor group g.
$\tilde{a}_{\{g,t\}}$	Source-standardised factor surprise, so groups are comparable.
λ	Fixed softmax temperature controlling concentration.

$\sum_g r_{g,t} = 1$

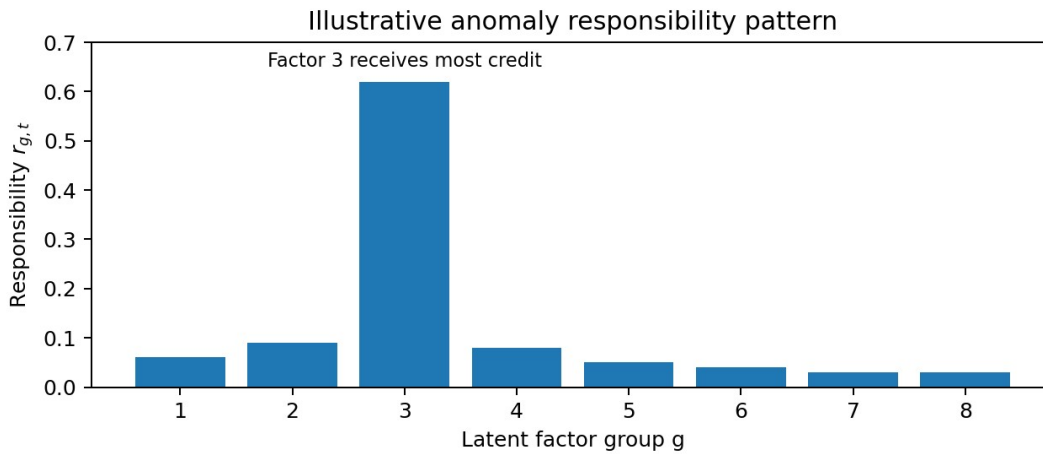
Responsibilities form a probability-like allocation across groups.

What the equation means

The softmax turns relative factor surprise into a write-credit vector. A factor does not have to mean a known semantic concept. It only needs to carry a stable share of the anomalous evidence.

Industrial example (illustrative)

For one defective item, group 3 may carry 62% of total responsibility while the other groups carry much less. This suggests that memory changes related to the current surprise should primarily target group 3.



12. C2: factor-addressed normal-memory update

This stage is tested only on queried samples verified as normal. The reasoning is: if the system thought the item was anomalous but the human says it is normal, then the surprising factor pattern may represent legitimate normal variation.

Equation (13). Global update control

$$\alpha_{g,t} = 1 \quad \forall g$$

Symbol	Meaning
$\alpha_{g,t}$	Effective update weight for factor group g.
1 for all groups	Every factor receives the current verified-normal observation equally.

What the equation means

This is the undifferentiated control. It mimics the idea of learning from the whole sample without asking which factor actually caused the false alarm.

Equation (14). Factor-addressed update

$$\alpha_{g,t} = r_{g,t}$$

Symbol	Meaning
$\alpha_{g,t}$	Effective contribution of the sample to factor group g .
$r_{g,t}$	Responsibility from Eq. (12).

What the equation means

Each factor is updated in proportion to its responsibility. Highly responsible factors learn more; unrelated factors learn little.

Industrial example (illustrative)

If a harmless reflection mainly activates factor group 3, the verified-normal update should expand the normal distribution of group 3 much more than groups 1, 2, 4, etc.

Equation (15). Collateral posterior drift

$$C_t = \sum_{g \neq g^*} D_{\text{KL}}\left(B_{t+1}^{(g)} \parallel B_t^{(g)}\right), \quad g^* = \arg \max_g r_{g,t}$$

Symbol	Meaning
g^*	The factor group with maximum responsibility.
$B_t^{(g)}$	Bayesian normal belief for factor group g before the update.
$B_{t+1}^{(g)}$	Belief after the update.
D_{KL}	Amount of distributional change caused by the update.
C_t	Total change in all factors except the primary credited group.

What the equation means

C_t directly measures unintended memory change. C_2 is supported only if factor-addressed writing reduces this collateral change without materially damaging detection.

Industrial example (illustrative)

If factor 3 caused the false alarm, a good selective update changes factor 3 strongly but leaves factors 1,2,4,... nearly unchanged. A global update changes all of them and therefore has higher collateral drift.

13. C3: source-enriched defective feedback

For defective queried samples, normality is not expanded. Instead, the scaffold asks whether a human defect-source label can stabilise the relationship between a recurring defect manifestation and the latent responsibility pattern that tends to express it.

Equation (16). Running source centroid in responsibility space

$$\mathbf{q}_{d,t} = \frac{1}{n_{d,t}} \sum_{i \leq t: d_i=d, a_i=query} \mathbf{r}_i$$

Symbol	Meaning
$\mathbf{q}_{\{d,t\}}$	Average responsibility pattern associated with source label d.
$n_{\{d,t\}}$	Number of queried examples of source d observed so far.
$\mathbf{r}_i$	Responsibility vector for a previous queried item.
$d_i=d$	Only queried items with the same source label contribute.

What the equation means

The source label does not name a latent factor. Instead, repeated human-labelled examples reveal a characteristic pattern over factors.

Industrial example (illustrative)

If several human-labelled "surface-mark" defects consistently produce high responsibility in groups 2 and 5, $\mathbf{q}_d$ becomes a stable two-group pattern even though neither group is called "surface mark".

Equation (17). Combining current machine responsibility with source memory

$$\alpha_t^{\text{src}} = \eta_s \mathbf{r}_t + (1 - \eta_s) \mathbf{q}_{d,t}, \quad \alpha_t^{\text{bin}} = \mathbf{r}_t$$

Symbol	Meaning
α_t^{src}	Credit vector used when the source/aspect label is available.
η_s	Fixed mixing coefficient selected on source validation.
$\mathbf{r}_t$	Current sample responsibility.
$\mathbf{q}_{\{d,t\}}$	Historical centroid for the current defect source.
α_t^{bin}	Binary-only control: uses current $\mathbf{r}_t$ without source memory.

What the equation means

The source-enriched condition blends evidence from the current image with what past human-labelled cases of the same source usually looked like in responsibility space.

Industrial example (illustrative)

A current defective sample has noisy responsibilities because of pose or lighting. The source centroid regularises that one-off pattern toward the recurrent pattern associated with the human-provided defect type.

14. Factor-level correction**Equation (18). Bounded residual correction**

$$\delta_{g,t+1} = \text{clip}(\delta_{g,t} + \eta_{\delta} \omega_t \alpha_{g,t}, -\delta_{\max}, \delta_{\max})$$

Symbol	Meaning
$\delta_{g,t}$	Current additive correction for factor group g.
η_{δ}	Correction learning rate.
ω_t	-1 for a verified false positive, +1 for a verified false negative.
$\alpha_{g,t}$	Credit assigned to factor g.
$\delta_{\max}$	Maximum magnitude of correction.
clip	Bounds the correction for stability.

What the equation means

A queried mistake can adjust factor-level anomaly evidence in the direction indicated by the error. False positives should suppress credited anomaly evidence; false negatives should increase it.

Industrial example (illustrative)

If a recurrent harmless visual pattern causes false positives and factor 3 receives most credit, δ_3 is pushed downward. If a subtle defect is repeatedly missed and factor 5 receives credit, δ_5 is pushed upward.

Explanatory expansion - How the additive correction is applied

$$\tilde{a}_{g,t} = a_{g,t} + \delta_{g,t}$$

Why this is included

The scaffold states that $\delta_{g,t}$ is a bounded additive residual applied to factor evidence. The simple expression above makes that sentence explicit. It is an explanatory implementation relation, not a new claim.

15. Negative controls: proving the effect is meaningful

- Shuffle source labels among queried defects while preserving source frequencies. If performance is unchanged, the benefit is not coming from meaningful source information.
- Randomly permute which latent coordinates belong to which factor groups after representation learning. If addressability is unchanged, the factor grouping is probably not carrying stable structure.

16. C1: what "factor addressability" means operationally

Important scope boundary

Addressability does NOT require "factor 3 = scratch" or "factor 5 = deformation". It requires a repeatable mapping from anomalous observations to a small subset/pattern of factors that correlates with independent defect-type and/or spatial evidence.

Test	Procedure	What success means
Source separability	Build q_d centroids from queried source labels and classify later queried defects by nearest centroid.	Different defect manifestations occupy distinguishable regions of responsibility space.
Responsibility stability	Compare cosine similarity within the same source versus between different sources.	Repeated examples of the same source have more similar responsibility patterns.
Spatial consistency	Use the max-responsibility factor's patch map and compare it offline with the ground-truth anomaly mask.	The credited factor is expressed where the physical anomaly is annotated.

17. C4: deterministic versus Bayesian governance

The deterministic comparator must use the same representation, same query sequence, same support set, same credit weights, and same correction update. The only difference is uncertainty modelling.

Condition	Normal-state model	Expected difference
Deterministic	Running mean and variance; Gaussian plug-in likelihood.	Acts as if point estimates are sufficiently known.
Bayesian	NIG posterior + Student-t posterior predictive.	Carries uncertainty from small effective sample sizes into anomaly surprise and state-change diagnostics.

Where C4 should matter most

Four-shot initialization, rare normal modes, random support variation, small numbers of queried normals, controlled label noise, memory pressure, or long gaps before recurrence. If deterministic and Bayesian models behave the same there, the extra Bayesian machinery is not justified.

18. Querying: only after C1-C4 survive

For C1-C4, replay the same saved query indices for every condition. This prevents a changed detector from creating a different supervision trajectory. Only C5 allows each method to choose its own queries.

Equation (19). One-step value of querying

$$V_t^b = P_t(y = 0)D_{\text{KL}}(B_t^+(z_t) \parallel B_t) + P_t(y = 1)U_t^{\text{attr}}$$

Symbol	Meaning
V_t^b	Approximate value assigned to querying item t.
$P_t(y=0)$	Current probability that the item is normal.
$B_t^+(z_t)$	Hypothetical Bayesian normal belief after a verified-normal update.
$D_{\text{KL}}(B_t^+ \parallel B_t)$	How much useful normal-memory state would change if the item were confirmed normal.
$P_t(y=1)$	Current probability that the item is defective.
U_t^{attr}	Attribution uncertainty if the item is defective.

What the equation means

The query value has two branches. If the item is likely normal, a query may be useful because it would change the Bayesian normal state. If the item is likely defective, a query may be useful because it can resolve uncertain attribution/source information. This is a tractable one-step approximation, not exact full-system information gain.

Industrial example (illustrative)

A borderline product may be 70% likely normal and, if confirmed normal, would strongly expand a poorly known factor distribution. Another product may be 95% likely defective but have very diffuse factor responsibility; a query can still be valuable because it teaches where the defect evidence belongs.

Equation (20). Attribution uncertainty

$$U_t^{\text{attr}} = - \sum_g r_{g,t} \log r_{g,t}$$

Symbol	Meaning
U_t^{attr}	Entropy of the responsibility vector.
$r_{\{g,t\}}$	Responsibility assigned to factor g.
High entropy	Responsibility is spread across many factors, so attribution is uncertain.
Low entropy	One or a few factors dominate, so attribution is clearer.

What the equation means

This term distinguishes "I am unsure whether it is defective" from "I am sure it is defective but unsure which

latent aspect carries the evidence."

19. Metrics that correspond directly to the claims

Equation (22). Write selectivity

$$S_t^{\text{write}} = \frac{D_{\text{KL}}\left(B_{t+1}^{(g^*)} \parallel B_t^{(g^*)}\right)}{\sum_g D_{\text{KL}}\left(B_{t+1}^{(g)} \parallel B_t^{(g)}\right) + \epsilon}$$

Symbol	Meaning
S_t^{write}	Fraction of total posterior change assigned to the most credited factor.
g^*	Top credited factor.
epsilon	Small constant preventing division by zero.

What the equation means

A selective memory write should concentrate change in the responsible factor instead of spreading it across all factors. Higher S_{write} is evidence of targeted writing, but it must be interpreted together with collateral drift and detection performance.

Equation (23). Queries required to reach an operating target

$$B(F^*)$$

Symbol	Meaning
F^*	Pre-specified operating point/function, e.g. a target accuracy/FNR combination.
$B(F^*)$	Number of human queries needed to reach F^* .

What the equation means

C5 is about supervision efficiency, not merely terminal accuracy. A better controller should reach the agreed operating point with fewer queries or dominate more of the performance-query frontier.

Claim	Primary mechanism metrics	Operating metrics retained
C1	Macro source accuracy/F1; within-minus-between responsibility similarity; pixel AUROC/AU-PRO.	Accuracy, AUROC, FPR, FNR.
C2	Collateral posterior drift C_t ; write selectivity	No material operating-performance loss.

	S_write.	
C3	Future attribution quality; performance per query; shuffled-source negative control.	Accuracy/FPR/FNR at identical query indices.
C4	Posterior-predictive NLL/calibration; seed variability; drift under label noise.	Detection competitiveness.
C5	Area under performance-query curve; B(F*).	Realised query rate and operating performance.

Part III - How One Online Step Works

20. Prequential update order

"Prequential" means predict first, then learn. This ordering is essential: the current hidden label may not influence the current recorded prediction.

14. Load the current target image x_t and cached frozen DINO patch features $h_{\{t,p\}}$.
15. Encode each patch into latent factors $z_{\{t,p\}}$ using the already-frozen factorizer.
16. Compute Bayesian posterior-predictive surprise $a_{\{g,t,p\}}$, aggregate to $a_{\{g,t\}}$, and derive responsibility r_t .
17. Compute the image anomaly score s_t and the current normal/defect decision.
18. Compute the query value under the active rule.
19. Log all pre-update outputs: score, decision, responsibility, query action, and state summary.
20. If no query is selected, do not reveal y_t and do not perform feedback-driven updates.
21. If queried, reveal y_t . Reveal d_t only in the source-enriched experimental condition.
22. If $y_t=0$ (verified normal), update the normal Bayesian state using the experimental update rule: global or factor-addressed.
23. If permitted by the condition, update the source centroid and/or correction state.
24. Update the query budget state and continue to $t+1$.

21. Walk-through example: one false positive and one defective query

21.1 False positive that teaches normality

25. The detector scores the product as anomalous because factor 3 has high surprise.
26. The query rule asks a human. The human says the product is normal.
27. Global control: all eight factors receive $\alpha=1$ and all normal posteriors move.
28. Factor-addressed condition: factor 3 receives most of the update because r_3 is largest; unrelated factors receive small fractional updates.
29. Later similar normal products should produce less surprise in factor 3, while the rest of the normal model is less disturbed.
30. C2 is tested by comparing collateral drift and future operating behaviour under identical queried items.

21.2 Defective sample with source-enriched feedback

31. The detector is queried and the human confirms "defective" plus a source/aspect label d_t .
32. Normal belief is NOT expanded because $y_t=1$.
33. The current responsibility pattern r_t is combined with the historical centroid $q_{\{d,t\}}$.
34. The resulting credit α_t^{src} directs the factor-level correction more consistently than r_t alone.
35. Across later queried defects of the same source, the responsibility centroid becomes more stable.
36. C3 is supported only if correct source labels outperform binary-only and shuffled-source controls at the same query indices.

Part IV - Detailed Research Process

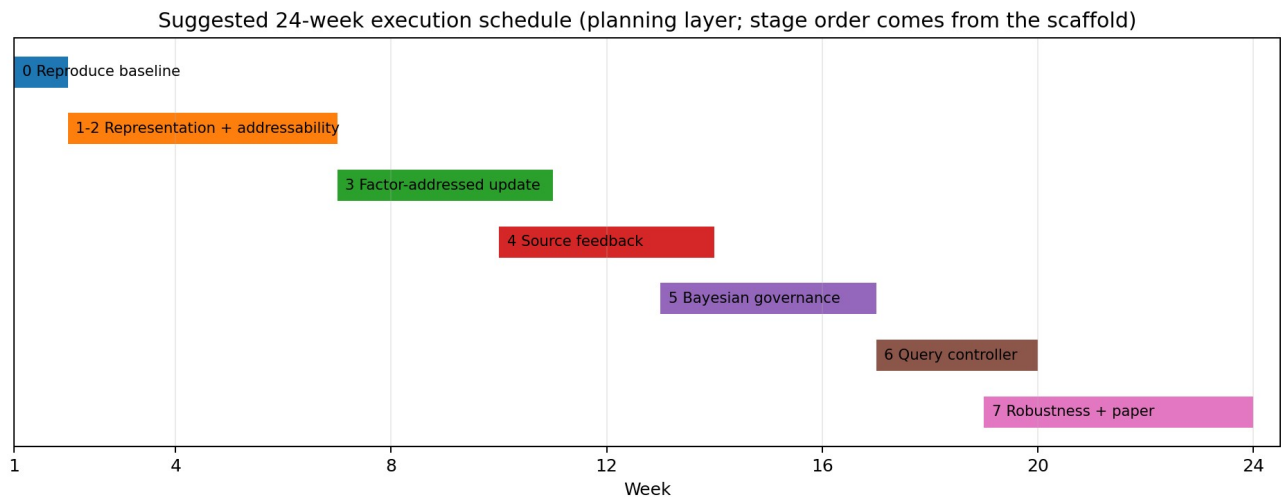
22. The scaffold's stage gates

Stage	Change introduced	Matched comparison	Scientific decision
0	Reproduce first-paper reference.	Current implementation + published protocol.	Establish numerical continuity.
1	Feature-space VAE/beta-VAE.	R1 vs R_beta; target representation frozen.	Does factorisation preserve anomaly information?
2	Addressability diagnostics.	R1 vs R_beta on separability, stability, masks.	If no measurable addressability appears, reject C1 and stop escalation.
3	Factor-addressed normal update.	All-factor vs r_t-weighted; fixed-query replay.	Does C2 reduce collateral adaptation?
4	Source-enriched defect feedback.	Binary vs (y,d); fixed-query replay + shuffled-source control.	Does C3 extract reusable value from richer human information?
5	Bayesian normal state.	Deterministic Gaussian vs NIG posterior predictive; fixed replay.	Does C4 improve sparse/noisy stability?
6	End-to-end query controller.	Decision uncertainty vs V_t^b at matched realised budget.	Does C5 improve supervision efficiency?
7	Robustness.	Support seeds, stream seeds, label/source noise, memory pressure.	Define claim boundaries rather than find one favourable setting.

23. Suggested 24-week schedule

Planning note

The scaffold specifies the stage order and stop/go logic but not durations. The schedule below is a six-month planning layer aligned with the scaffold and your available research window.



Period	Main work	Goal / exit condition
Weeks 1-2	Stage 0: reproduce first-paper Real-IAD logs; freeze folds, configs, query replay indices, and logging schema.	Numerical continuity established; reproducible baseline ready.

Weeks 2-5	Stage 1: build DINO feature cache; source-fold linear projection; train R1 and R_beta; freeze target-time modules.	R1/R_beta reconstruct DINO features and preserve useful anomaly ranking.
Weeks 5-7	Stage 2: compute factor surprise, responsibilities, source separability, stability, and spatial maps.	C1 gate. Proceed only if R_beta shows measurable addressability beyond R1 without unacceptable detection loss.
Weeks 7-11	Stage 3: implement NIG state, deterministic comparator, fractional updates, global vs factor-addressed replay, KL diagnostics.	C2 gate: lower collateral drift with competitive detection.
Weeks 10-14	Stage 4: source centroids, source-enriched credit, factor correction, shuffled-source and factor-permutation controls.	C3 gate: correct source labels improve future attribution/operation over controls.
Weeks 13-17	Stage 5: deterministic vs Bayesian governance under four-shot, random supports, sparse queried normals, and controlled noise.	C4 gate: Bayesian advantage appears where uncertainty should matter.
Weeks 17-20	Stage 6: implement V_t^b query controller; compare with decision uncertainty and random at matched budgets.	C5 gate: better frontier or lower $B(F^*)$.
Weeks 19-24	Stage 7 + writing: robustness, cross-fold aggregation, bootstrap CIs, figures, ablations, paper draft.	Claim boundaries documented; submission-ready evidence package.

24. What to implement in each period

Period A - Experimental foundation

- Freeze category folds, support seeds, stream seeds, and data-access rules.
- Generate fixed-query replay indices from the first-paper controller for mechanism isolation.
- Create machine-readable configs and per-item log schema before adding new methods.
- Goal: every later comparison is reproducible and differs in only one declared mechanism.

Period B - Representation feasibility

- Cache frozen DINO patch descriptors once.
- Train R1 and R_beta only on source-category normal features.
- Measure reconstruction in feature space and anomaly-ranking retention.
- Compute addressability diagnostics before implementing sophisticated memory.
- Goal: establish whether factor addressability is empirically real enough to justify the rest of the study.

Period C - Selective memory mechanism

- Implement the NIG normal state and deterministic Gaussian comparator.
- Implement factor surprise, responsibility, fractional NIG update, and KL state-drift logging.
- Replay identical verified normals under global and factor-addressed updates.
- Goal: show the credit vector changes where memory is written, not merely the final AUROC.

Period D - Richer human evidence

- Use Real-IAD defect type d_t as the first source/aspect label.
- Maintain q_d centroids only from queried defects.
- Add factor-level correction and the shuffled-source / factor-permutation controls.
- Goal: show that source-enriched human information has reusable value beyond binary labels.

Period E - Bayesian governance

- Hold representation, queries, credit, support, and correction fixed.
- Change only deterministic point estimates versus NIG posterior predictive uncertainty.
- Stress low-data/noisy regimes where Bayesian uncertainty has a reason to matter.
- Goal: justify Bayesian machinery empirically rather than because it is mathematically elegant.

Period F - Query efficiency and robustness

- Only now allow each method to choose its own queries.
- Compare decision uncertainty, the proposed V_t^b rule, and random querying at matched realised budgets.
- Run support/stream/noise/memory-pressure robustness and aggregate over category-seed units.
- Goal: demonstrate greater deployment value per human intervention.

25. Stop/go rules

Claim	Proceed if...	If it fails...
C1	R_{β} improves addressability over R_1 while retaining non-inferior anomaly utility.	Do not add Bayesian querying to hide the failure; revise or abandon the factor-addressability thesis.
C2	Factor-addressed update lowers collateral drift with no material operating loss.	Responsibility is not useful as memory-address selection; revise credit/update rule.
C3	Correct source labels beat binary-only and shuffled-source feedback at identical queries.	Richer human labels are not producing reusable value in this representation.
C4	Bayesian model improves a pre-specified calibration/stability endpoint in sparse/noisy regimes.	Use the simpler deterministic state; Bayesian machinery is not justified.
C5	Information-aware query rule improves frontier area or reaches F^* with fewer queries.	Keep the established memory mechanism but reject the new query claim.

26. Software modules and what each one is allowed to know

Module	Input -> output	Target-time rule
Frozen feature extractor	image -> cached patch descriptors	No trainable target-time state.
Factorizer	patch descriptor -> $(\mu_z, \log \sigma_z^2, z)$	Frozen during target stream.
Normal state	support/update statistics -> posterior predictives + KL drift	Uses only permitted support and queried-normal updates.
Attribution	factor surprise -> r_t + spatial factor maps	No access to hidden labels.
Source memory	queried source label + r_t -> q_d	No access to unqueried source labels.
Correction	queried outcome + factor credit -> δ_t	Only queried feedback changes it.
Query controller	current score/responsibility/state -> classify/query	Must act before current hidden label is revealed.
Evaluator/logger	pre-update outputs + offline GT for evaluation	Offline ground truth is marked unavailable to the online algorithm.

27. Immediate deliverables before full coding

37. Frozen Real-IAD category-level fold assignment.
38. Data-access table defining what is visible at source training, target support, target stream, and queried feedback.

39. R1 and R_beta representation configurations.
40. NIG hyperparameter-estimation procedure based only on source categories.
41. Fixed-query replay indices for mechanism experiments.
42. Per-item and per-run logging schema.
43. Predeclared primary metrics and non-inferiority tolerance derived from source-fold variability only.
44. Machine-readable C1-C5 claim-decision table stored with experiment configurations.

28. What should NOT be implemented initially

- Raw-pixel VAE reconstruction.
- Bayesian neural networks or Bayesian backbone weights.
- Dirichlet-process/nonparametric memories.
- Causal manufacturing-process graphs.
- Free-text human explanations.
- Multi-view fusion on Real-IAD.
- Factor-specific operating thresholds.
- Learned end-to-end attribution classifier.
- Semantic claims that an unsupervised latent factor is a physical root cause.

Reason for exclusion

None of these is necessary to test the central chain: factor-addressable frozen representation -> credit assignment -> selective memory update -> uncertainty-aware governance. Adding them early would make a negative result difficult to interpret.

29. What would make the final study convincing?

45. The factorised frozen representation retains adequate anomaly ranking.
46. Factor responsibilities correspond reproducibly to independent defect-type and/or spatial evidence.
47. Credit-weighted normal updates produce less unrelated posterior change.
48. Correct source-enriched feedback helps future attribution or operating behaviour more than binary and shuffled-source feedback.
49. Bayesian posterior prediction improves stability/calibration when evidence is scarce or noisy.
50. Information-aware querying converts the improved state representation into a better performance-human-intervention frontier.

Most important interpretation

The study does not need every stage to increase AUROC. Each claim has its own mechanism-specific observable. The research is strongest when it can explain why a component is useful, not merely when a large final system has a better terminal number.

30. Final mental model

Remember the chain

Frozen DINO gives a stable visual substrate. The beta-VAE is only a candidate way to make the substrate addressable. Bayesian normality tells us how surprising each factor is under sparse normal evidence. Responsibility converts surprise into write credit. Human feedback decides whether normality should expand or whether a source/correction memory should be updated. Querying is added last, once the memory mechanism is trustworthy.

Source basis and interpretation policy

This guide is grounded in the uploaded "Minimum Implementation Scaffold for Attribution-Aware Bayesian Memory Governance, Version 0.1". The 24-week schedule and the explicit fractional NIG / Student-t / corrected-evidence equations are explanatory planning or implementation expansions added to make the scaffold easier to understand; they are not additional claims from the source document.